

Ravi Chepuri

612-860-7253 – rchepuri@umd.edu – ravi-chepuri.github.io

Education

University of Maryland, College Park (UMD)

Ph.D., Physics

Advisor: Michelle Girvan

College Park, MD

Expected 2027

Northwestern University (NU)

B.A., Physics, Math, and Integrated Science, *Magna Cum Laude* (GPA: 3.97)

2022

Honors thesis: “A Scalable Architecture Towards Complex Quantum Communication Networks”

Evanston, IL

Research Experience

Ph.D. Candidate, Dept. of Physics, UMD

2022—Present

Advisor: Michelle Girvan

- Developed approaches based on dynamical systems and statistical physics to characterize how node heterogeneity influences the dynamics of reservoir computers, a recurrent neural network architecture with fixed weights.
- Implemented large scale computational methods on high-performance computing platforms to simulate and analyze nonlinear dynamical systems, including reservoir-based models.
- Applied these methods to model brain activity using reservoir computing, integrating neurophysiological data (EEG, fMRI) with biophysically grounded dynamical descriptions to advance toward a digital twin of brain dynamics.

Undergraduate Researcher, Dept. of Physics and Astronomy, NU

2020—2022

Advisor: István Kovács

- Developed a theoretical framework for constructing quantum communication networks on lattices of interacting quantum spins.
- Performed numerical simulations to evaluate network properties arising from spin-mediated interactions.
- Demonstrated that these quantum networks can exhibit substantial structural complexity.

Publications

1. **Chepuri, R.** & Girvan, M. Heterogeneity in reservoir computers: a reduced-order dynamical analysis. *In preparation*.

2. **Chepuri, R.**, Amzalag, D., Antonsen, T. M. & Girvan, M. Hybridizing traditional and next-generation reservoir computing to accurately and efficiently forecast dynamical systems. *Chaos: An Interdisciplinary Journal of Nonlinear Science* 34, 063114 (2024).
3. **Chepuri, R.** & Kovács, I. A. Complex quantum network models from spin clusters. *Communications Physics* 6, 271 (2023).

Presentations

1. **Chepuri, R.** & Girvan, M. “The role of heterogeneity in reservoir computing dynamics.” International Physics of Living Systems conference, 2025 Jul 28 - Aug 1, Urbana, IL. (*poster presentation*)
2. **Chepuri, R.** & Girvan, M. “Exploring the role of heterogeneity in reservoir computing dynamics.” Northeast regional conference for complex systems, 2025 Apr 9-11, Binghamton, NY. (*oral presentation*)
3. **Chepuri, R.** & Kovács, I.A. “Complex quantum networks on a lattice of spins.” APS March Meeting, 2023 Mar 5-10, Las Vegas, NV. (*oral presentation*)
4. **Chepuri, R.** & Kovács, I.A. “Can we design a complex quantum network?” International School and Conference on Network Science, 2020 Sep 21-25, Rome, Italy (online). (*poster presentation*)
5. **Chepuri, R.** “A scalable quantum network architecture.” Northwestern Undergraduate Research and Arts Exposition, 2020 May 27-28, online. (*poster presentation*)

Awards

NSF GRFP Honorable Mention	2024
Graduate Student Seminar prize, UMD (best seminar talk of spring 2024)	2024
Oliver Marcy Scholar, NU (top 3 undergrads in natural sciences)	2022
Fletcher Undergraduate Research Grant Award, NU (best STEM undergrad research by faculty committee review)	2022
Phi Kappa Beta junior inductee, NU (top ~2% of class)	2021
Sigma Pi Sigma junior inductee, NU (physics honor society)	2021
University-nominated candidate for the Barry Goldwater Scholarship, NU	2021
Outstanding Sophomore/Junior in Physics and Astronomy, NU	2020, 2021
DAAD RISE Germany scholarship (rough REU equivalent; canceled, Covid-19)	2020
Dean’s List, NU (12/12 academic quarters)	2018 – 2022

Professional Development

Complex Systems Summer School, Santa Fe Institute, Santa Fe, NM

2024

Teaching Experience

Teaching Assistant, Dept. of Physics, UMD

Fall 2022 (1 semester)

Course: Fundamentals of Physics I. Led introductory physics labs and discussion sections (~35 students). Lectured for weekly 15-minute review sessions. Developed quiz and discussion questions.

Teaching Assistant, Dept. of Physics and Astronomy, NU

Summer 2022 (1 quarter)

Course: College Physics I. Led discussion sections (~25 students), held weekly office hours, proctored and graded exams

Teaching Assistant, Dept. of Mathematics, NU

Fall 2021—Winter 2022 (2 quarters)

Courses: Multivariable Differential Calculus. Led discussion sections (~25 students), held weekly office hours, proctored and graded exams

Mentorship Experience

Graduate Mentor, TREND-REU, UMD

2023

Mentored undergraduate in summer research project in nonlinear dynamics. Led to a publication in *Chaos* journal.

Graduate Mentor, GRAD-MAP, UMD

2023

Mentored undergraduate in a research challenge question as part of a workshop for undergraduates at HBCUs and minority serving institutions. Designed and created a pedagogical Python-based challenge question to create and analyze a numerical simulation of the Kuramoto model.

Peer Mentor, Office of Undergraduate Research, NU

2020-2021

Led weekly small group workshops for undergraduates interested in starting research. Facilitated student discussions about diversity and inclusion in STEM. Held individual check-ins with students supported by the Summer Undergraduate Research Grant.